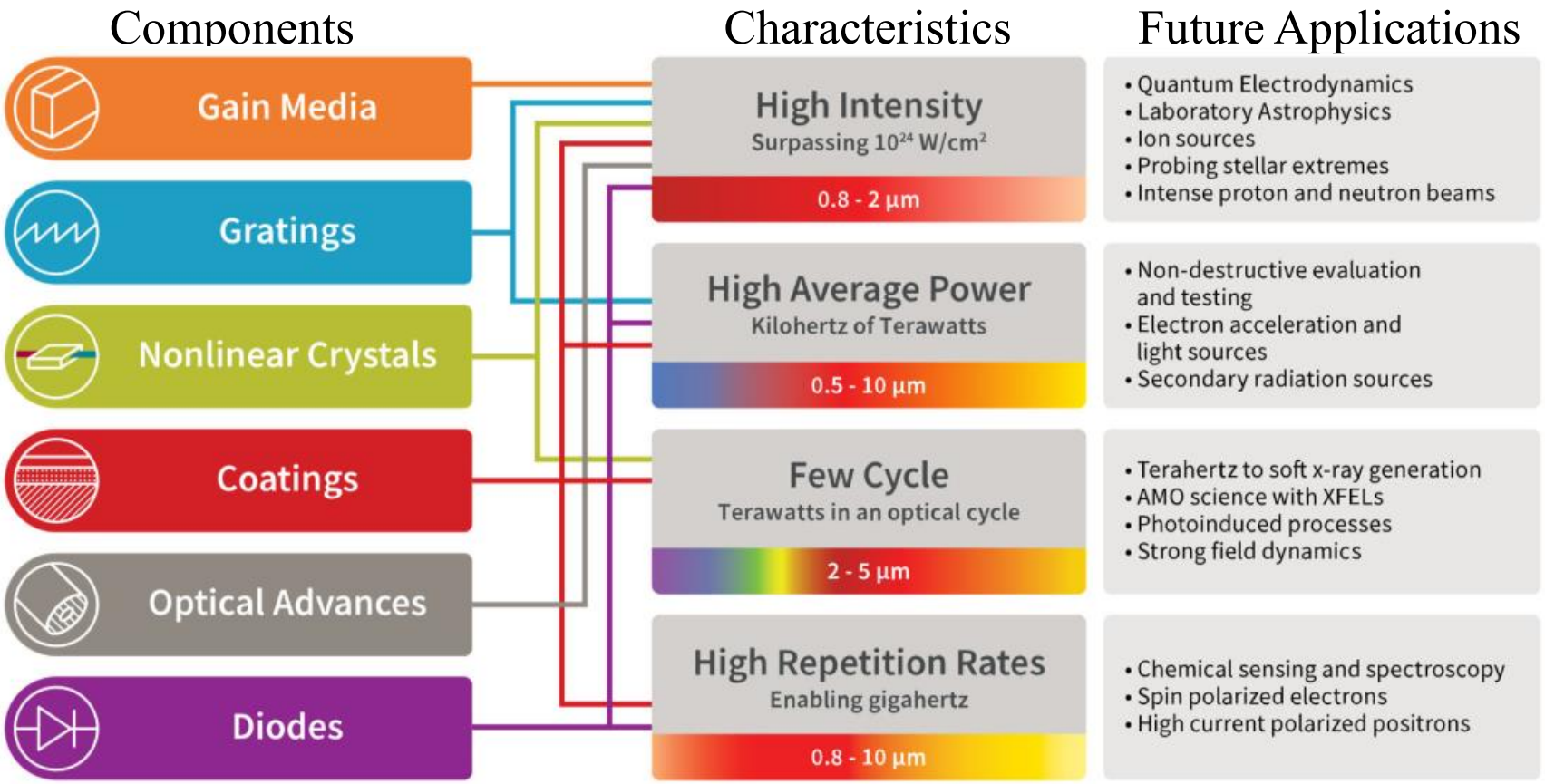
Electron Dynamics of Femtosecond Laser Induced Plasma inside Multilayer Dielectric High Reflectors Studied Using Particle-in-cell (PIC) Approach

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[4] *The Ohio State University, Department of Physics, Columbus, OH 43210*

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Motivation

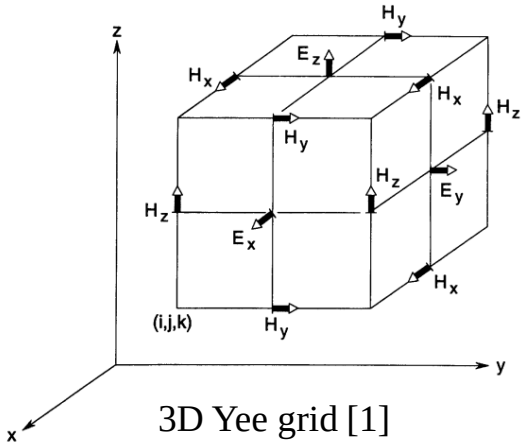


Fundamental technological advances and their enabling technologies [1]

Mirrors are identified as the weakest parts in the CPA system. Therefore, it is crucial to increase the Laser Induced Damage Threshold (LIDT) of these components.

[0] Matthias F. Kling, REPORT OF THE BASIC RESEARCH NEEDS WORKSHOP ON LASER TECHNOLOGY, 2023

FDTD



3D Yee grid [1]

Maxwell equations

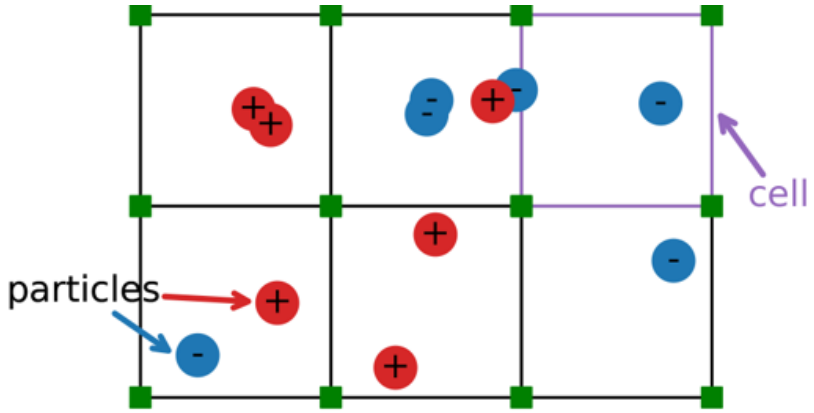
$$\nabla \times E = \frac{-\partial B}{\partial t}$$

$$\nabla \times H = \frac{\partial D}{\partial t} + J$$

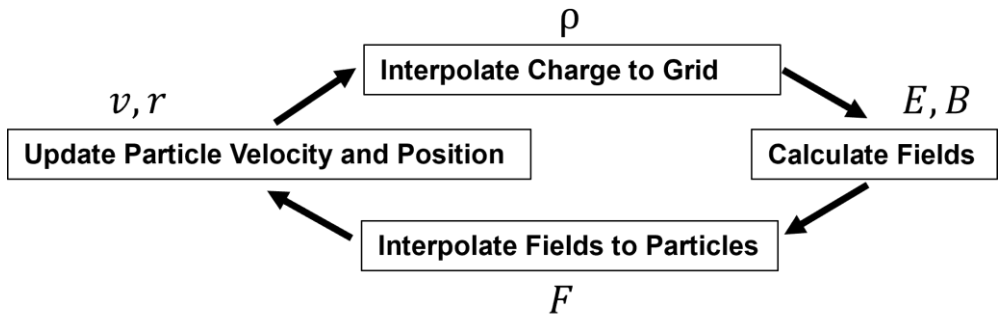
$$D = \epsilon E$$

$$B = \mu H$$

PIC



Sketch of a 2D PIC simulation grid



Both Finite Difference Time Domain (FDTD) and PIC are based on mesh grid approaches for solving electromagnetic fields, but FDTD cannot reveal the particle dynamics, while Particle In Cell (PIC) can.

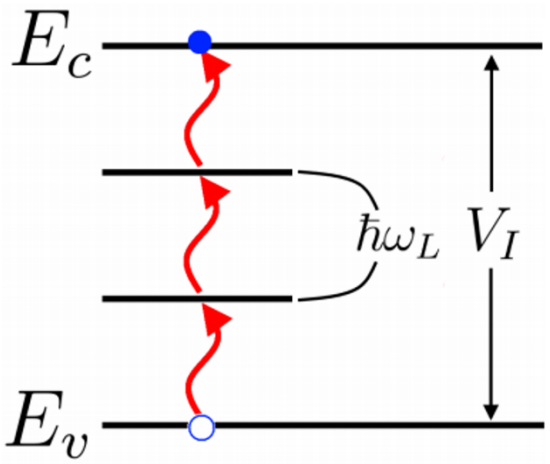
[1] Kane Yee, "Numerical solution of initial boundary value problems involving maxwell's equations in isotropic media," in *IEEE Transactions on Antennas and Propagation*, vol. 14, no. 3, pp. 302-307, May 1966, doi: 10.1109/TAP.1966.1138693.

Keldysh Parameter [2]

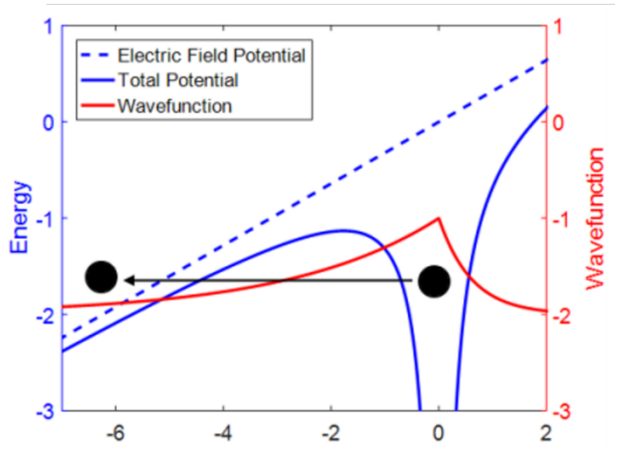
$$\gamma = \frac{\omega\sqrt{m_e W_0}}{eE}$$

W_0: Ionization energy
E: External electric field
ω : Laser frequency

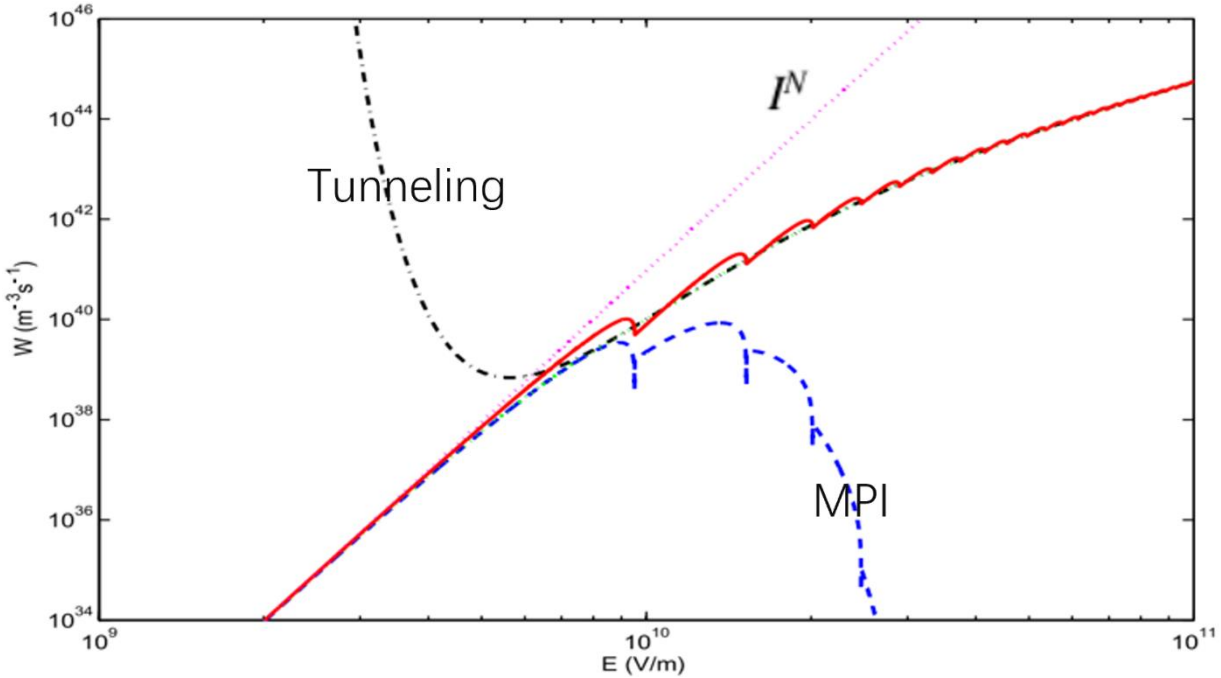
Multi-photon/Tunneling Ionization Cartoon [3]



$\gamma \gg 1$



$\gamma \ll 1$



Strong Field Ionization Rates Based on Different Models (wavelength 800 nm, bandgap 6.5 eV) [4]

[2] L. V. Keldysh, "Ionization in the field of a strong electromagnetic wave," J. Exp. Theor.Phys., vol. 20, no. 5, pp. 1307–1314, 1965
[3] A. RUSSELL, "Modeling ultrashort pulse laser damage with the particle in cell method," Ph.D. dissertation, 2019.
[4] P. Balling and J. Schou, "Femtosecond-laser ablation dynamics of dielectrics: Basics and applications for thin films," Reports on Progress in Physics, vol. 76, no. 3, p. 036 502, Feb.2013. DOI: 10.1088/0034-4885/76/3/036502.

Method Modified PIC

Laser-Plasma Interactions [5]

EPOCH
CFSA
WARWICK
An open source PIC code for high energy density physics

github.com/Warwick-Plasma/epoch

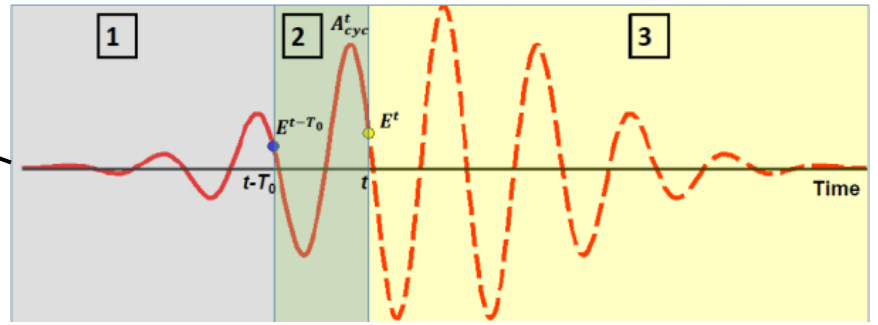
Optical Material Properties

Keldysh Photoionization

Impact Ionization

Implemented refractive index into the field solver

Cycle-averaged approach

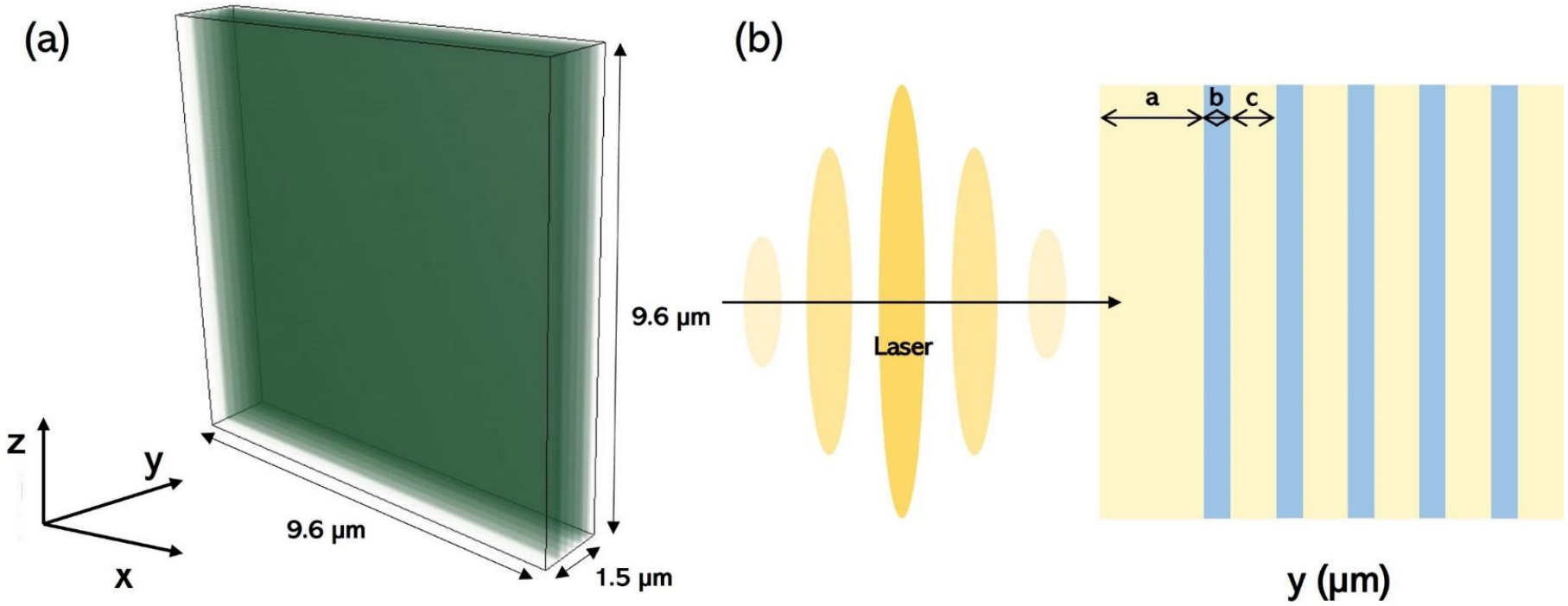


A sketch of time profile of electric field at a fixed space point.[6]

[5] T D Arber, K Bennett, C S Brady, A Lawrence-Douglas, M G Ramsay, N J Sircombe, P Gillies, R G Evans, H Schmitz, A R Bell, "Contemporary particle-in-cell approach to laser-plasma modelling," Plasma Physics and Controlled Fusion, 2015.
[6] S. Zhang, C. Menoni, V. Gruzdev, and E. Chowdhury, "Ultrafast laser material damage simulation—a new look at an old problem," Nanomaterials 12, 1259 (2022)

3D Simulation Set-up Source and Resolution

Source
Wavelength: 800 nm
Duration: 7 fs
$1/e^2$ Waist: 4.65 μm
Single Gaussian Pulse
Normal Incident



(a) 3D simulation window. (b) 2D schematic pulse and quarter-wave mirror with a silica protective layer

Resolution
40 nm (x, z): 15 nm (y)
1000 Macroparticle per cell
Loaded 3.9 billion SiO_2 and 2.1 billion HfO_2

Wall time
~10-15 hours with 5 nodes

Materials Properties [7-11]

	m^*_e	m^*_h	W_0(eV)	n	Mass Density (g/cc)	Dissociation Energy Density (kJ/cm^3)	Melting Energy Density (kJ/cm^3)	Boiling Energy Density (kJ/cm^3)
a-SiO_2	8	0.6	9	1.4765	2.20	54-68	5.7	28.7-35.6
a-HfO_2	1.07	1.12	5.7	2.0212	9.68	48-53	10.7	49.9

⋆ Spherically averaged effective mass, heat capacity and dissociation energy of a-HfO_2 are based on m-HfO_2

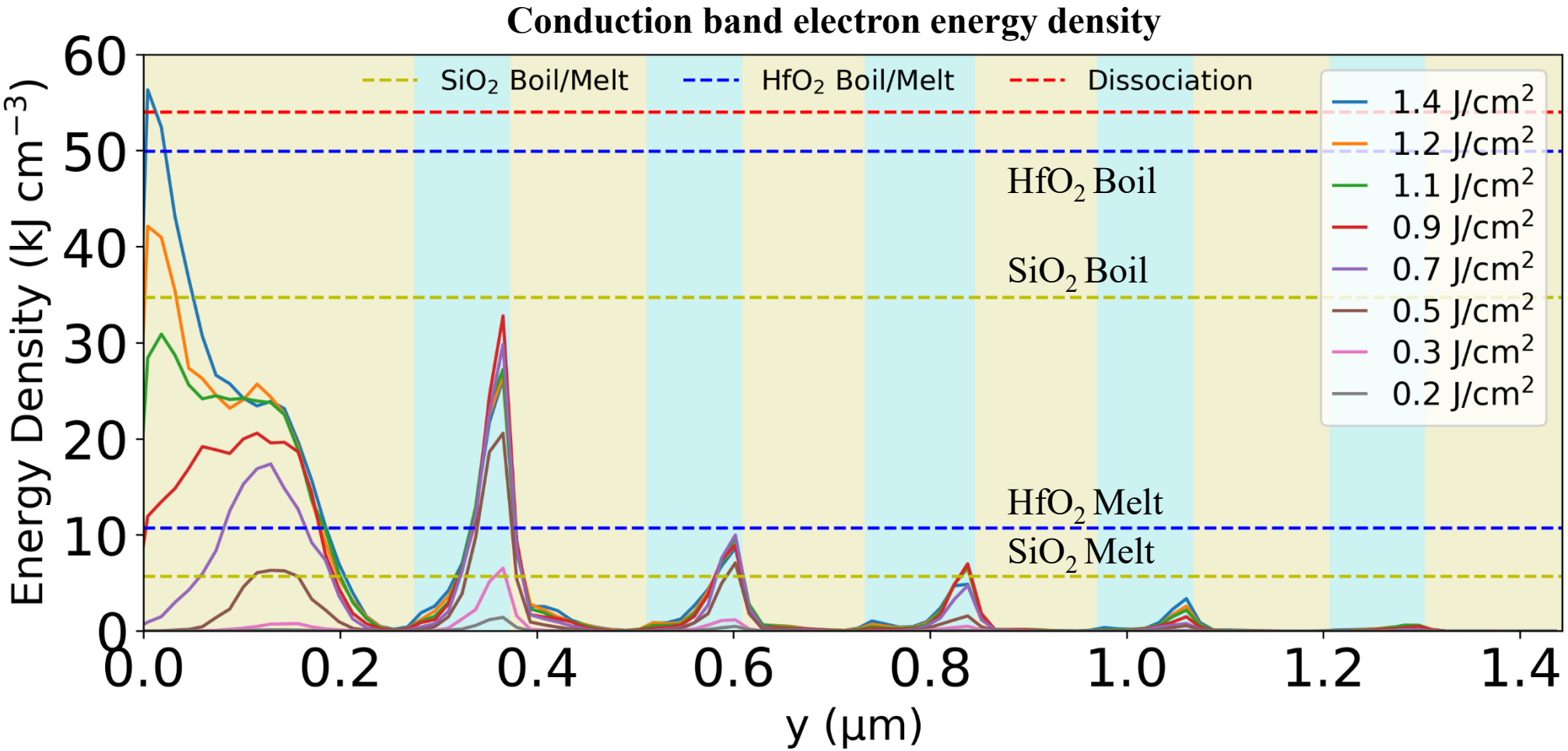
[7] Zhang, S.; Tripepi, M.; AlShafey, A.; Talisa, N.; Nguyen, H.T.; Reagan, B.A.; Sistrunk, E.; Gibson, D.J.; Alessi, D.A.; Chowdhury, E.A. Femtosecond damage experiments and modeling of broadband mid-infrared dielectric diffraction gratings. Opt. Express 2021, 29, 39983–39999.
[8] J. J. Low, N. H. Paulson, M. D'Mello, and M. Stan, "Thermodynamics ofmonoclinic and tetragonal hafnium dioxide (HfO2) at ambient pressure,"Calphad 72 (2020), 10.1016/j.calphad.2020.102210.
[9] Wang C, Zinkevich M, and Aldinger F. The Zirconia–Hafnia System: DTA Measurements and Ther-modynamic Calculations. Journal of the American Ceramic Society 2006;89:3751–8.
[10] J. Zhao, J. Sullivan, J. Zayac, and T. D. Bennett, "Structural modification ofsilica glass by laser scanning," Journal of Applied Physics 95, 5475–5482(2004), https://doi.org/10.1063/1.1703832.
[11] M. Grehn, T. Seuthe, M. Höfner, N. Griga, C. Theiss, A. Mermillod-Blondin, M. Eberstein, H. Eichler, and J. Bonse, "Femtosecond-laser in-duced ablation of silicate glasses and the intrinsic dissociation energy," Op-tical Materials Express 4, 689–700 (2014).

LIDT Criteria Energy Density

Dissociation: Slightly less than 1.4 J/cm^2

Boil: Between 1.1-1.2 J/cm^2

Melt: Between 0.3-0.5 J/cm^2

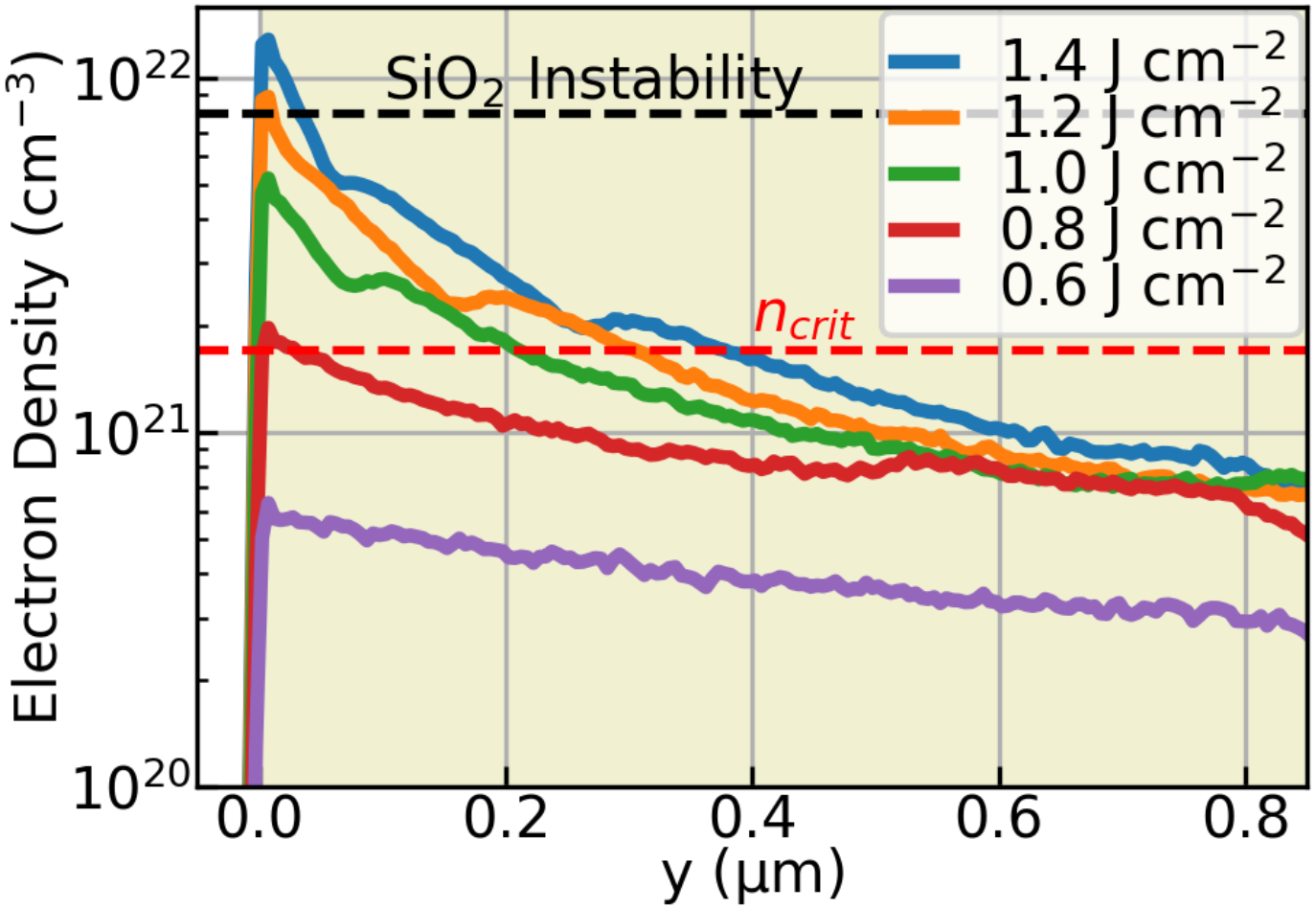


LIDT Criteria Electron Density

Instability: Slightly less than 1.2 J/cm^2

Experiment: 1.18 J/cm^2 [12]

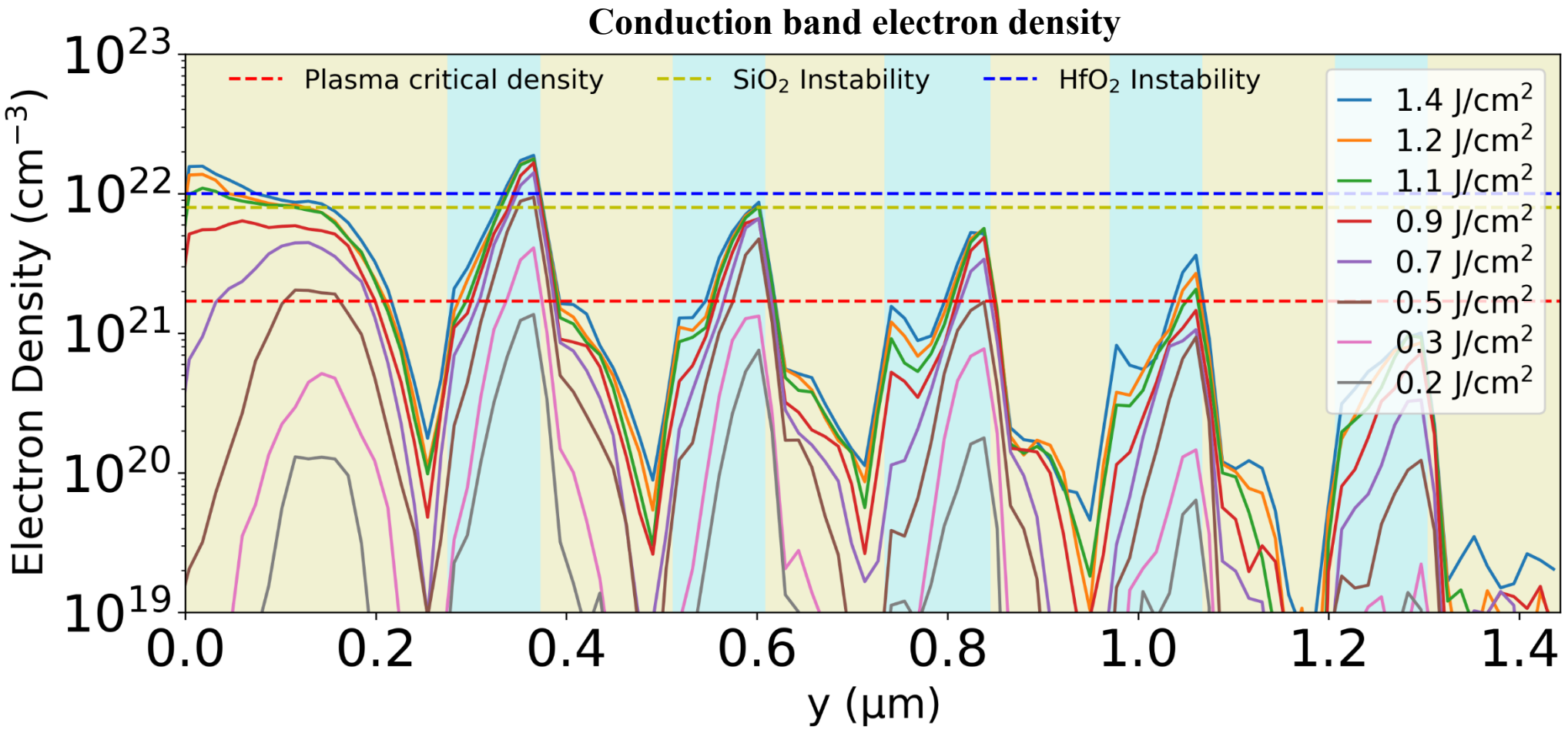
When the conduction band electron density reaches about 9% of valence band electron density, the elastic shear constant will become negative and the lattice becomes unstable, which then leads directly to a very rapid melting of the crystal structure [13].



[12] B. Chimier et al., "Damage and ablation thresholds of fused-silica in fem-tosecond regime," Phys. Rev. B 84, 094104 (2011).
[13] P. Stampfli and K. H. Bennemann, "Theory for the instability of the diamond structure of Si, Ge, and C induced by a dense electron-hole plasma,"Phys. Rev. B 42, 7163–7173 (1990).

Instability:
About 0.5 J/cm²

Plasma Critical:
About 0.2 J/cm²

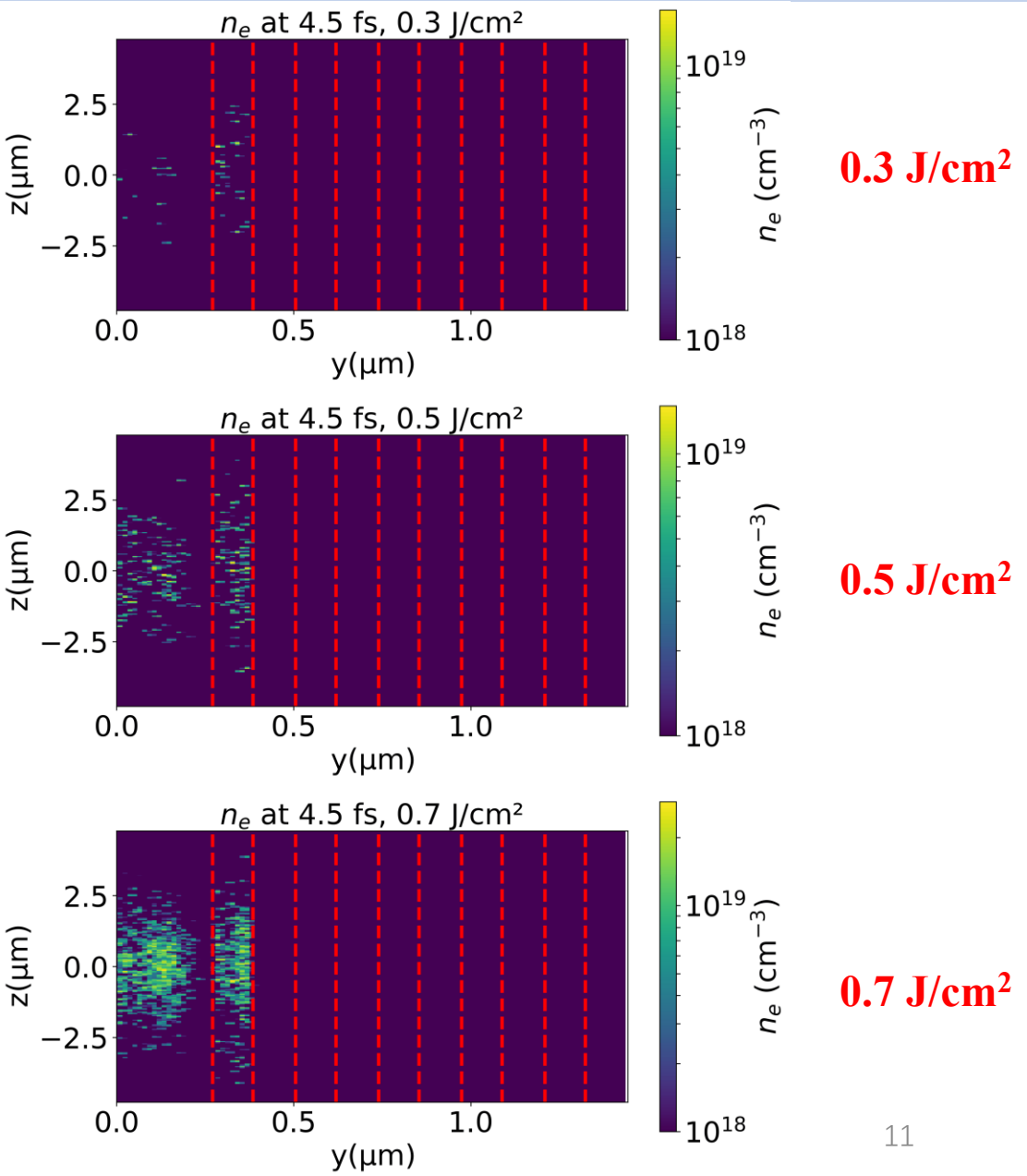


Electron Density Dynamics

There is an expansion of the local peak electron density to the surface at higher fluence.

Implies

Plasma Generation



0.3 J/cm²

0.5 J/cm²

0.7 J/cm²

Electron Density Dynamics

Plasma generation at the center of the top SiO_2 layer

↓

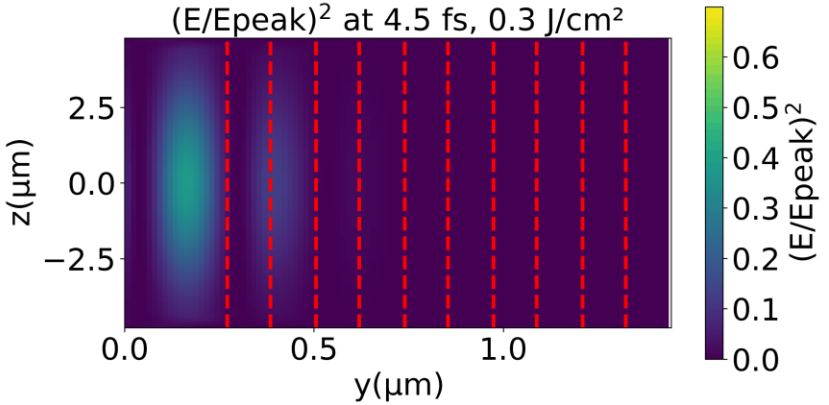
The plasma alters the optical properties and increases the reflection

↓

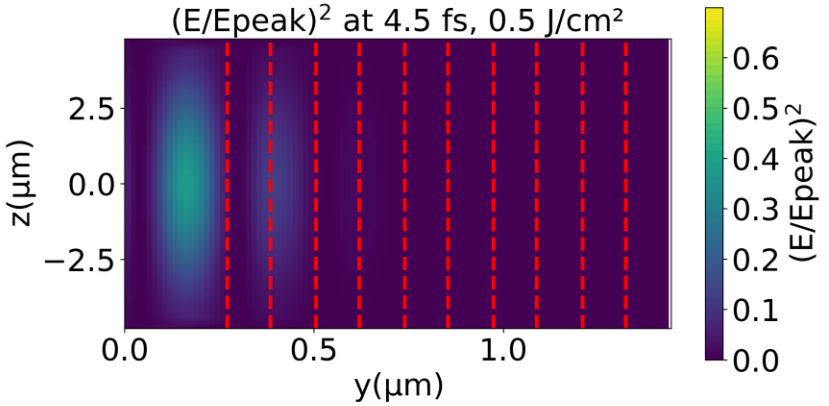
The resonant electric field pattern is changed and thus shifts the constructive peak to the surface

↓

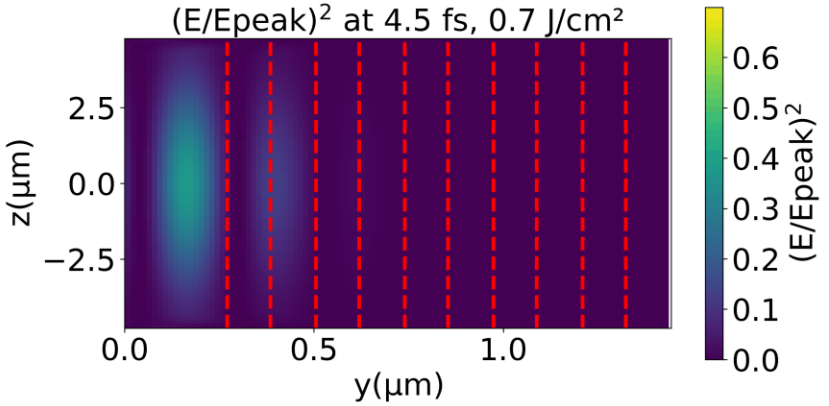
The ionization is then enhanced due to the high intensity



0.3 J/cm²

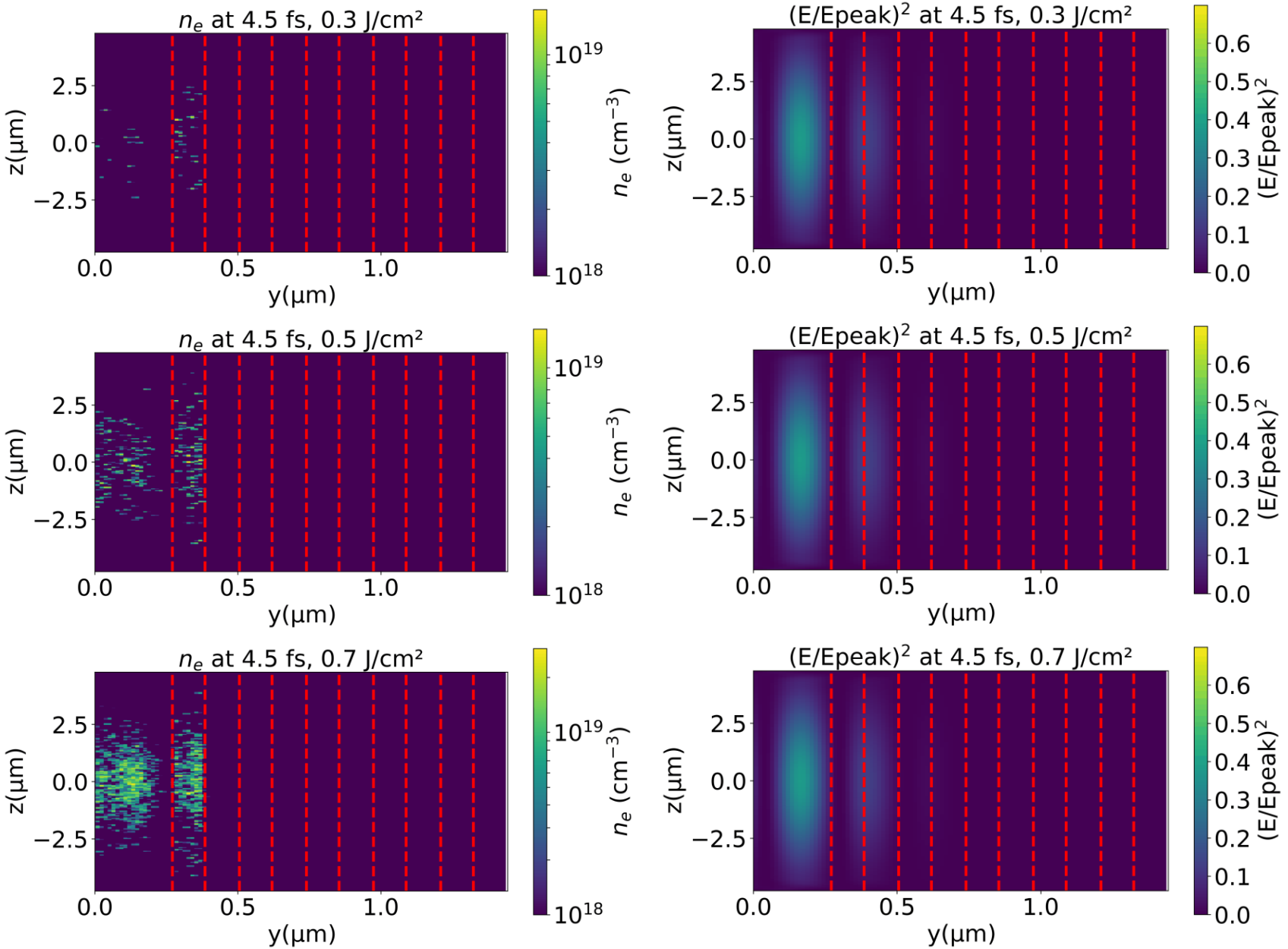


0.5 J/cm²

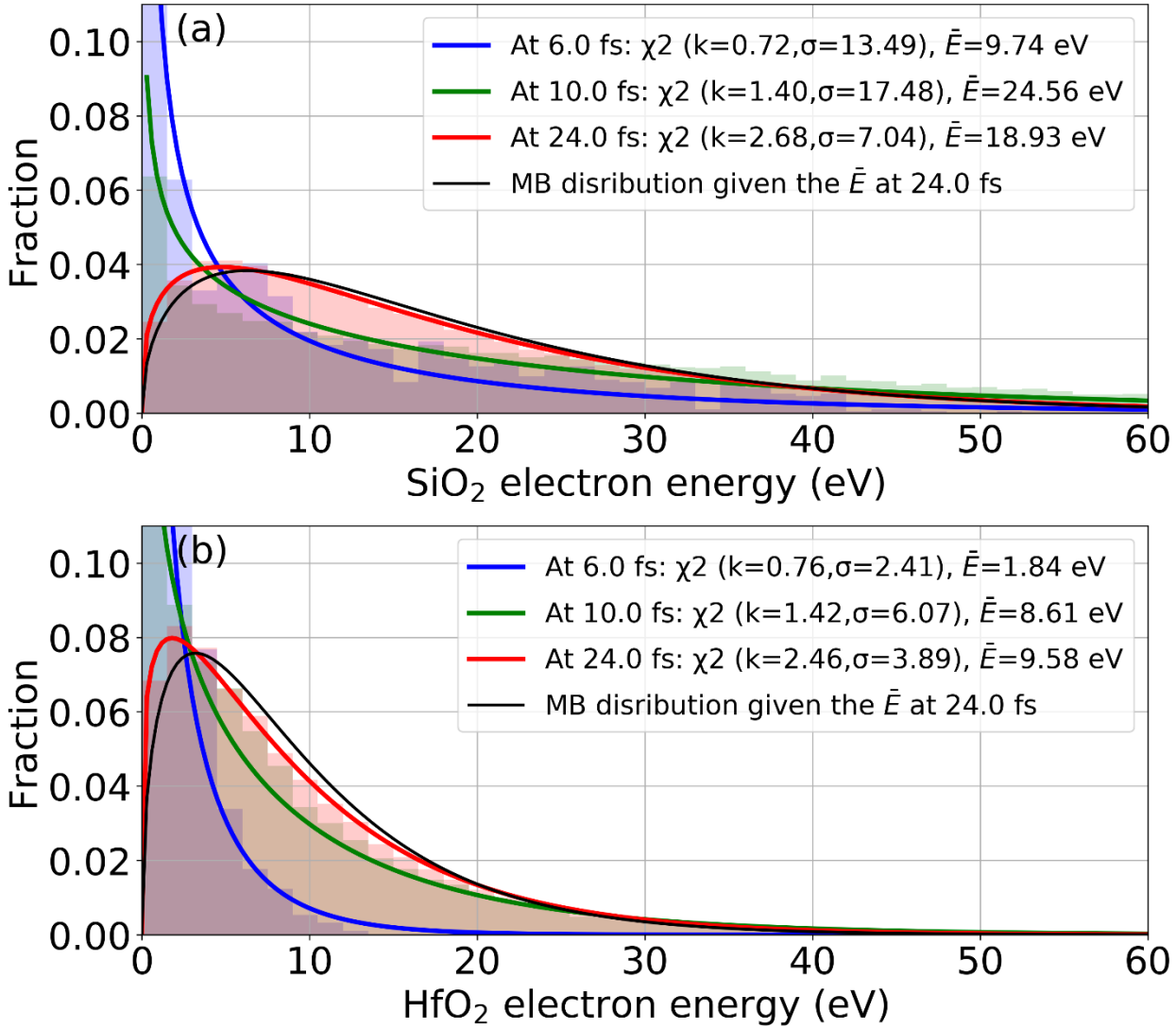


0.7 J/cm²

Electron Density Dynamics



Electron Thermalization



The energy histogram for the electrons in the (a) first SiO_2 layer and (b) HfO_2 layer at 0.7 J cm^{-2}

$$f(E_e; k, \sigma) = \frac{1}{\sigma^2} \frac{E_e^{\left(\frac{k}{2}-1\right)} e^{-\frac{E_e}{2\sigma^2}}}{2^{\frac{k}{2}}\Gamma\left(\frac{k}{2}\right)} \quad [13]$$

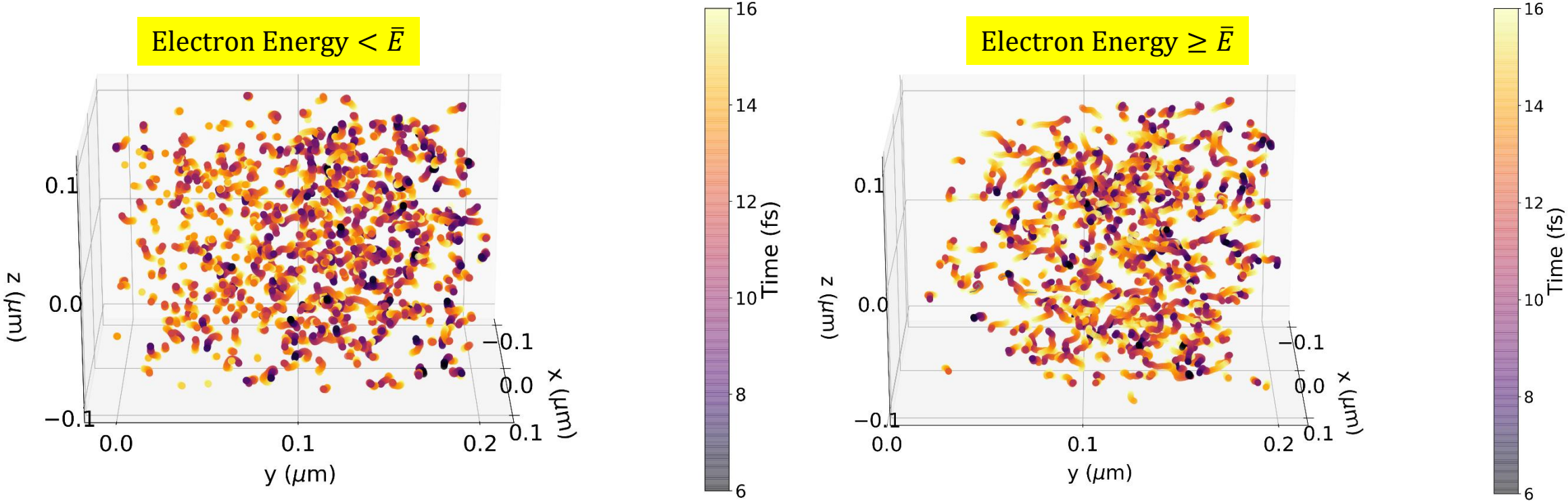
$$E_e: electron\ kinetic\ energy$$

When k = 3, the Chi^2 distribution becomes the MB distribution

[14] P. Virtanen et al., "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods* 17, 261–272 (2020).

Electron Displacement

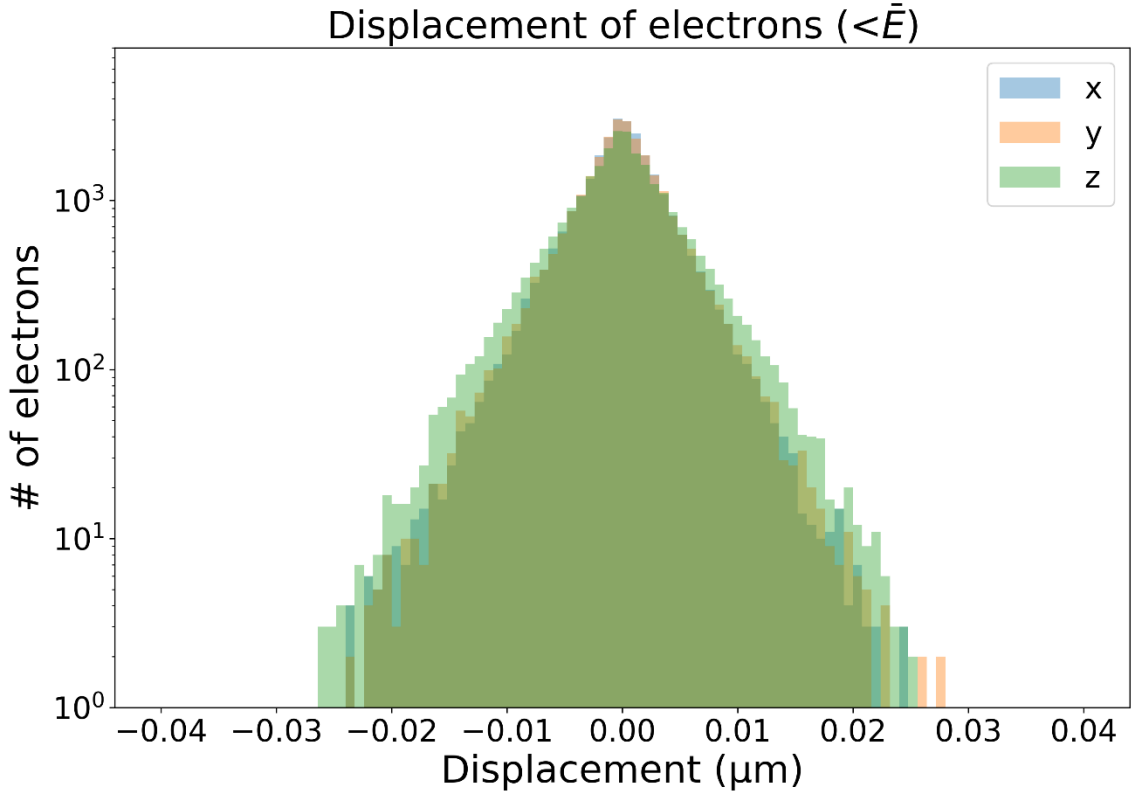
The track of conduction band electrons near air-SiO_2 interface from 6 fs to 16 fs at 0.7 J/cm^2

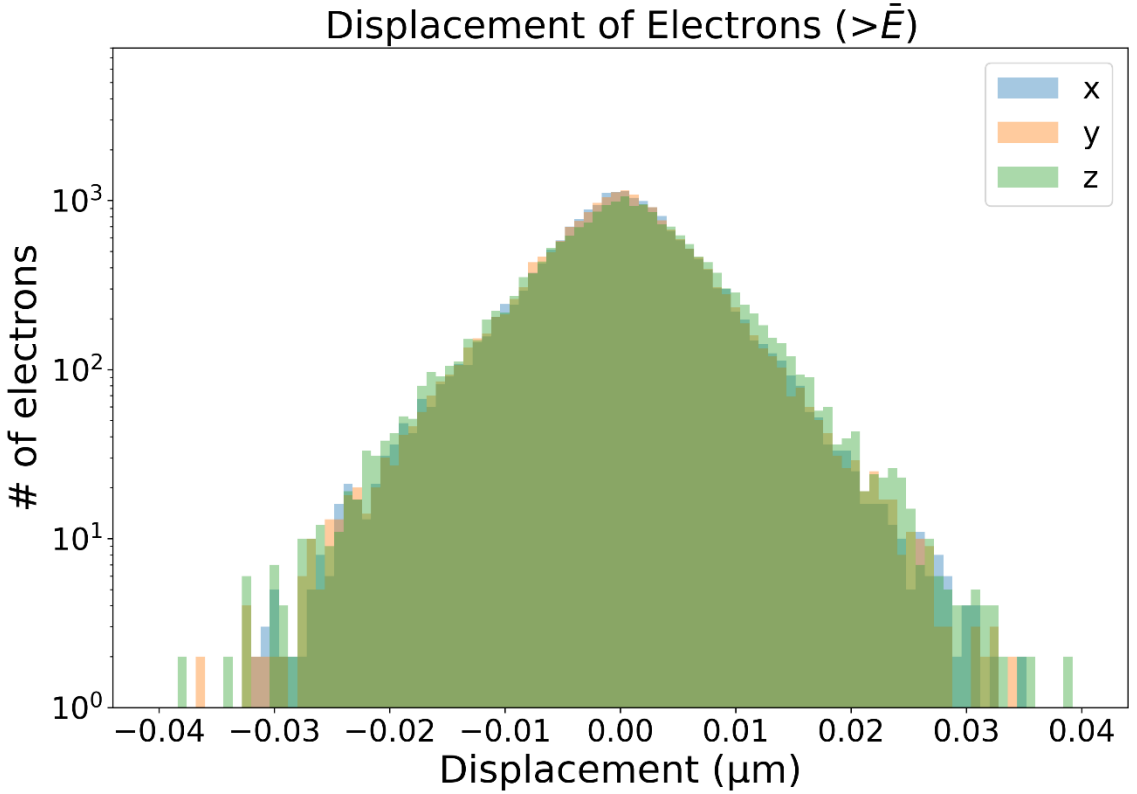


The conduction band electrons near the air-SiO_2 interface are born more lately, while the electrons near the center of top SiO_2 layer are born earlier

Electron Displacement

The polarization is along z





Summary:

1. Implemented Keldysh photoionization model and spatially varying refractive index in EPOCH
2. Different criteria give different predictions, and instability electron density predicts LIDT at about 0.5 J/cm^2
3. Showed the screening effects of the plasma
4. Revealed thermalization and polarization of the electrons

Outlook:

1. Incorporate impact ionization model
2. Inject oblique source and apply to gratings

All the data and results presented here are part of manuscript in preparation:

Smith, J. R., Su, Z., Zhang, S., Varin, C., Gruzdev, V. E., & Chowdhury, E. A. (2024). *A Fully Three-Dimensional Kinetic Particle-In-Cell Framework for Modeling Laser-Dielectric Interactions: Few-Cycle Pulse Damage*.

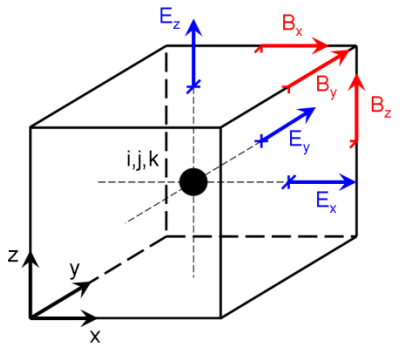
Thank you

Bibliography

[0] Matthias F. Kling, REPORT OF THE BASIC RESEARCH NEEDS WORKSHOP ON LASER TECHNOLOGY, 2023
[1] Kane Yee, "Numerical solution of initial boundary value problems involving maxwell's equations in isotropic media," in *IEEE Transactions on Antennas and Propagation*, vol. 14, no. 3, pp. 302-307, May 1966, doi: 10.1109/TAP.1966.1138693.
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[10] J. Zhao, J. Sullivan, J. Zayac, and T. D. Bennett, "Structural modification ofsilica glass by laser scanning," Journal of Applied Physics 95, 5475–5482(2004), https://doi.org/10.1063/1.1703832.
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[14] P. Virtanen et al., "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," *Nature Methods* 17, 261–272 (2020).
[15] Z. Su, J. R. Smith, and E. Chowdhury, "Electron Dynamics of Femtosecond Laser-Induced Plasma Inside Multilayer Dielectric High Reflectors Studied Using Particle-in-Cell (PIC) Approach," in *CLEO 2024*, Technical Digest Series (Optica Publishing Group, 2024), paper SF3H.5.

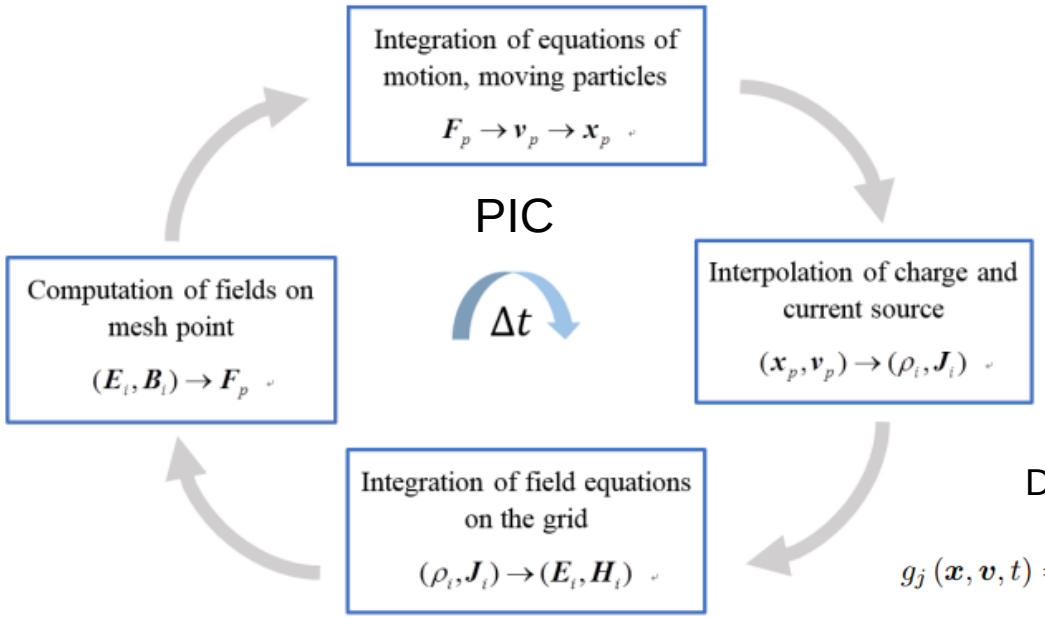
Back-up

Method Particle In Cell (PIC)



$$\begin{cases} \dfrac{d\boldsymbol{r}}{dt} = \boldsymbol{v} \\ \dfrac{d\boldsymbol{v}}{dt} = \dfrac{q}{\gamma m}(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) \end{cases}$$

Particle solvers



$$\nabla \cdot \boldsymbol{E} = \frac{\rho(\boldsymbol{x}, t)}{\epsilon_0},$$

$$\nabla \cdot \boldsymbol{B} = 0,$$

$$\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t},$$

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J}(\boldsymbol{x}, t) + \frac{1}{c^2}\frac{\partial \boldsymbol{E}}{\partial t}.$$

Maxwell equations

Distribution function

$$g_j(\boldsymbol{x}, \boldsymbol{v}, t) = \sum_{i=1}^{N_j} [\delta(\boldsymbol{x} - \boldsymbol{r}_i(t))\, \delta(\boldsymbol{v} - \boldsymbol{V}_i(t))]$$

$$\rho(\boldsymbol{x}, t) = \sum_j \left(q_j \int g_j(\boldsymbol{x}, \boldsymbol{v}, t)\, \mathrm{d}\boldsymbol{v} \right)$$

$$\boldsymbol{J}(\boldsymbol{x}, t) = \sum_j \left(q_j \int \boldsymbol{v} g_j(\boldsymbol{x}, \boldsymbol{v}, t)\, \mathrm{d}\boldsymbol{v} \right)$$